

- Q.32 If 15m^3 volume of an oil weight 75 KN, calculate
- a) Specific Weight
 - b) Mass density
 - c) Specific Gravity
- Q.33 Describe the term compressibility. What are its SI unit?
- Q.34 Explain the terms potential energy, kinetic energy, pressure energy and total head.
- Q.35 Explain in detail working of hydraulic brake with neat diagram.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the construction and operation of hydraulic accumulator with the help of a neat diagram.
- Q.37 What do you mean by intensity of pressure and pressure head of liquid. Why we measure pressure in term of head? Give its conversion and Units.
- Q.38 Explain the construction and working of reaction turbine with a neat diagram.

4th Sem / Mechatronics Subject:- Hydraulic & Pneumatic Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 An ideal fluid is a fluid which is
- a) Compressible
 - b) Incompressible and viscous
 - c) Incompressible
 - d) None of the above
- Q.2 The density of water is maximum at
- a) 0 deg C
 - b) 4 deg C
 - c) 273 K
 - d) 32 F
- Q.3 The flow in a pipe is laminar if
- a) $Re=2500$
 - b) $Re=3000$
 - c) $Re>3000$
 - d) None of the above
- Q.4 Hydraulic gradient line is
- a) Always below the total energy line
 - b) Always about the total energy line
 - c) Is always parallel to bottom
 - d) None of the above
- Q.5 The magnitude of water hammer depends upon
- a) Length of pipe

- b) Time taken to close the valve
 - c) Elastic properties of pipe material
 - d) All of the above
- Q.6 Hydraulic accumulator store the energy of fluid in the form of _____
- a) Potential energy b) Kinetic energy
 - c) Pressure energy d) All of the above
- Q.7 Centrifugal pump is started with its delivery valve kept
- a) Fully open
 - b) Fully closed
 - c) Partially open or close
 - d) None of the above
- Q.8 An ideal hydraulic oil should have
- a) Low flammability b) Low volatility
 - c) Low density d) All of the above
- Q.9 Orifice meter is use to measure
- a) Discharge b) Average velocity
 - c) Pressure at a point d) Specific weight
- Q.10 Printers ink is a/an
- a) Newtonian fluid b) Non-newtonian fluid
 - c) Ideal plastic fluid d) Thyrotrophic fluid

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 When pressure is expressed in terms of height of liquid, it is called _____
- Q.12 Laminar flow are also known as _____

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- Q.13 The S.I unit of discharge is _____
- Q.14 Total head of liquid= _____ + _____ + _____ m of liquid.
- Q.15 A nozzle converts _____ of liquid into _____
- Q.16 Pitot tube is use to measure density true or false.
- Q.17 Define seals.
- Q.18 According to Chezy's formula , $v =$ _____
- Q.19 A nozzle convert pressure energy into kinetic energy (True/False)
- Q.20 Turbine is prime mover? (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between Laminar and turbulent flow.
- Q.22 List the various advantages of manometers.
- Q.23 Explain the working of centrifugal pump with neat sketch.
- Q.24 Drive and state the continuity equation.
- Q.25 Differentiate between centrifugal pump and reciprocating pump.
- Q.26 Explain the priming and its various method.
- Q.27 List the various common problems related to pneumatic system.
- Q.28 Name the various type of seals with their application.
- Q.29 Explain pelton wheel turbine with a neat diagram.
- Q.30 What do you mean by manometer list its various type and explain also.
- Q.31 Explain the following
- a) Absolute pressure b) Pitot tube

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